among the workers as a class. All that each one of them ‘cares’ about is her own genes. A worker might have ‘liked’ to have chaperoned her own mother, but she lacked the opportunity, not having been conceived in those days. A young queen on her mating flight is the sister of the present generation of workers, not the mother. Therefore they are on *her* side rather than on the side of the next generation of workers, who are merely their nieces. My head is now spinning, and it is high time to bring this topic to a close.

I have used the analogy of farming for what hymenopteran workers do to their mothers. The farm is a gene farm. The workers use their mother as a more efficient manufacturer of copies of their own genes than they would be themselves. The genes come off the production line in packages called reproductive individuals. This farming analogy should not be confused with a quite different sense in which the social insects may be said to farm. Social insects discovered, as man did long after, that settled cultivation of food can be more efficient than hunting and gathering.

For example, several species of ants in the New World, and, quite independently, termites in Africa, cultivate ‘fungus gardens’. The best known are the so-called parasol ants of South America. These are immensely successful. Single colonies with more than two million individuals have been found. Their nests consist of huge spreading underground complexes of passages and galleries going down to a depth of ten feet or more, made by the excavation of as much as 40 tons of soil. The underground chambers contain the fungus gardens. The ants deliberately sow fungus of a particular species in special compost beds which they prepare by chewing leaves into fragments. Instead of foraging directly for their own food, the workers forage for leaves to make compost. The ‘appetite’ of a colony of parasol ants for leaves is gargantuan. This makes them a major economic pest, but the leaves are not food for themselves but food for their fungi. The ants eventually harvest and eat the fungi and feed them to their brood. The fungi are more efficient at breaking down leaf material than the ants’ own stomachs would be, which is how the ants benefit by the arrangement. It is possible that the fungi benefit too, even though they are cropped: the ants propagate them more efficiently than their own spore dispersal mechanism might achieve. Furthermore, the ants ‘weed’ the fungus gardens, keeping them clear of alien species of fungi. By removing